Change your mind in proportion

Revise confidence according to the relevance and strength of new evidence.

Suggested teaching time: 30 min | Preparation: about 5 min

Introductory | Secondary and adult learners | Individual writing, pairs, and class discussion

Prerequisites: None

Students will be able to

- State whether new evidence should raise, lower, or barely change confidence.
- Explain why stronger relevant evidence deserves a larger update.
- Keep a conclusion qualified when important details are missing.

Prepare and gather

- One student handout per learner, or display the activities and use paper.
- Pens or pencils; no student devices required.
- Instructor guide with model answers, kept separate from the handout.
- Read the three scenarios and their model answers.
- Print the three-page handout or display one activity at a time.
- Explain that all scenarios and figures are invented for reasoning practice.

Teaching sequence

5 min	**Match evidence to the claim.** Learners complete the reading-app warm-up. Distinguish liking an app from learning more.
10 min	**Compare two lighting reports.** Pairs complete the lighting activity. Require a reason for both the direction and relative size of each update, without exact confidence percentages.
8 min	**Discuss strength and limits.** Use the model answer and prompts. Explain that stronger evidence can justify greater confidence while leaving questions about size and reliability.
7 min	**Revise a different belief.** Learners complete the reminder exit ticket independently. Assess evidence relevance, proportionate revision, and a useful missing detail.

Timings are suggested starting points, not measured classroom completion times.

Warm-up / Does liking an app prove learning?

Student task and instructor answer

Explain these words as needed

Confidence: How strongly you believe a claim is true.

Relevant evidence: Information that bears on the particular claim being evaluated.

Random assignment: Using chance to decide which group receives a change, helping make groups comparable.

Scenario

Invented scenario: You are unsure whether a reading app improves comprehension. A friend says its colors are attractive. A report then describes a fair comparison in which app users answer more comprehension questions correctly than a similar group using usual materials; no details about the size of the difference are supplied.

Student output: Write how each report should affect confidence and one remaining question.

1. Explain whether the comment about colors bears on comprehension.

2. Say whether the comparison should raise or lower confidence, relative to starting unsure.

3. Name one missing detail needed before making a strong claim.

Student success check: Have you changed confidence for the learning result while keeping its limits visible?

Model answer

The color comment should change confidence about comprehension very little. The stated fair comparison should raise confidence more because it measures comprehension directly. I would ask how large the difference was and how reliably it was measured before making a strong claim.

Why this meets the task: The answer gives more weight to relevant outcome evidence without inventing a precise effect or confidence number.

Main activity / What should two studies change?

Student task and instructor answer

Scenario

Invented scenario: You begin unsure whether brighter streetlights reduce theft reports. Study A is one resident saying a street feels safer. Study B assigns 40 similar streets at random to brighter or unchanged lighting, keeps patrol policy the same, and records the same theft measure before and after: reports fall more in the brighter-light group. No exact effect size or uncertainty estimate is supplied.

Student output: Write one update for each study, a comparison of their strength, and one missing detail you would seek.

1. For each study, say whether it should raise, lower, or leave your confidence roughly unchanged, and explain why.
2. Choose which should change confidence more, using the difference between personal impressions and a controlled comparison.
3. Name one missing detail you would seek before making a strong conclusion. Do not invent a precise percentage change in confidence.

Student success check: Have you updated more for the stronger relevant evidence while keeping uncertainty about the size and reliability of the effect?

Model answer

Study A should change confidence about theft reports very little: feeling safer is a different outcome and one person's impression. Study B should raise confidence more because random assignment and consistent measurement provide a fairer comparison of the claimed outcome. I would request the size of the difference and its uncertainty before making a strong claim.

Why this meets the task: It ties the direction and strength of the update to evidence quality and acknowledges that the scenario cannot justify an exact confidence number.

Exit ticket / Do reminders prevent missed deadlines?

Student task and instructor answer

Scenario

Invented scenario: You tentatively believe reminder messages reduce missed deadlines. Report A says one user likes the messages. Report B randomly assigns 200 students to reminders or no reminders for the same four-week course, uses the same deadline rules, and reports nearly equal missed-deadline rates. Exact rates and uncertainty estimates are not supplied.

Student output: Write how each report should change your confidence, which matters more, and one question still open.

1. Judge how much the approval report bears on missed deadlines.

2. State the direction of the update from the comparison and explain why it matters more.

3. Name a missing detail and explain why the report does not prove reminders can never help.

Student success check: Have you allowed contrary evidence to weaken the belief without turning one result into a universal verdict?

Model answer

Report A should change confidence very little: liking messages is not fewer missed deadlines. Report B should lower confidence in a meaningful benefit for this setting because it compares the claimed outcome under random assignment. I would ask for the exact difference and its uncertainty. Nearly equal reported rates do not prove no possible benefit in any course or group.

Why this meets the task: The answer responds to contrary evidence, weights the controlled comparison more strongly, and limits the conclusion to what the incomplete report supports.

Check understanding

Mark each criterion met or needs another attempt. Accept equivalent wording; assess the reasoning, not a memorized phrase.

- The direction of each update matches the evidence and the starting belief.
- The stronger update is justified by relevance and method rather than confidence of presentation.
- The conclusion names a meaningful uncertainty and avoids invented precision or universal certainty.

Discussion and adaptations

Use the supplied guidance to keep the lesson focused on its reasoning skill.

Discussion prompts

Does stronger evidence require complete certainty?

Listen for: No. It can justify a larger change while leaving uncertainty about the size, reliability, or reach of the result.

Why avoid exact confidence percentages here?

Listen for: The scenarios do not supply enough information to calculate them. Explain the direction and relative strength of the change instead of pretending to numerical precision.

Common confusions

Changing a belief means admitting all earlier reasoning was worthless. A reasonable starting view can change when relevant evidence improves. Revision can be a sign of careful thinking.

One strong report settles every setting. Check the outcome, groups, methods, and missing details before extending the conclusion.

Fairness requires keeping the same confidence. Relevant evidence should matter. Avoid both ignoring it and overreacting beyond what it shows.

Adapt the lesson

More support: Offer three direction labels: raise, lower, very little change. Require one reason naming the measured outcome before comparing the size of the updates. Expected example: the color comment changes confidence very little; the fair comprehension comparison raises it, with questions about effect size remaining.

More challenge: Ask: If five news stories repeat the same study, are they five independent confirmations? Expected response: no; they share one underlying result. Separate repetition from new evidence before increasing confidence further.

Next practice

https://xixteen.com/learners/?practice=belief_update#workbench